

SECTION INDEX: Power Screws & Screw Jacks Classification Index

Machine Element / Joint Type	Primary Design Parameter / Objective	Databook Section
Complete Design of Screw Jack (80 kN Load)	Complete design of Spindle (d_c, d_o, p), Nut (n, h, D_1, D_2), Handle & Cup (L, D, η), & Body ($D_5, D_6, D_7, H_{\text{body}}$)	Section 1
Differential Screw Jack Analysis	Analyze differential torque (T), efficiency (η), & max load capacity (W_{max})	Section 2
Standard Square Threads	IS: 4694 – 1968 Fine Series square thread dimensions table	Ref Table 17.1

Design Standard Note: All design procedures, friction torques, principal stress checks, nut shear stress equations, and differential screw formulas conform strictly to standard mechanical design practice (IS: 4694 – 1968 for Square Threads).

SECTION 1: Complete Design of a Screw Jack (80 kN Load)

System Parameters

Lift Load: $W = 80 \text{ kN} = 80000 \text{ N}$ | Max Lift Height: $H_1 = 400 \text{ mm} = 0.4 \text{ m}$

Screw Material: $\sigma_{ec} = 200 \text{ MPa} \Rightarrow$ Permissible $\sigma_c = 100 \text{ N/mm}^2$, $\tau_e = 120 \text{ MPa} \Rightarrow$ Permissible $\tau = 60 \text{ N/mm}^2$

Nut Material (Phosphor-Bronze): $\sigma_{t(\text{nut})} = 50 \text{ MPa}$, $\sigma_{c(\text{nut})} = 45 \text{ MPa}$, $\tau_{\text{nut}} = 40 \text{ MPa}$

Bearing Pressure: $p_b = 18 \text{ N/mm}^2$ | Friction: $\mu = \tan \varphi = 0.14$, Swivel Collar $\mu_1 = 0.14$

Design Protocol

Part 1: Design Screw Spindle (d_c, d_o, p, d , Torque T_1 , Principal Stresses $\sigma_{c(\text{max})}, \tau_{\text{max}}$, Buckling W_{cr})

Part 2: Design Nut (Threads n , Height h , Shear Stresses $\tau_{\text{screw}}, \tau_{\text{nut}}$, Diameters D_1, D_2 , Thickness t_1)

Part 3: Design Handle & Cup ($D_3, D_4, T_2, T, L, D, H, \eta$)

Part 4: Design Body Proportions ($D_5, D_6, D_7, t_2, t_3, H_{\text{body}}$)

1. Preliminary Core Diameter (d_c)

Direct compression under load $W = 80000 \text{ N}$

$$\begin{aligned}
 W &= \frac{\pi}{4} (d_c)^2 \cdot \sigma_c \\
 80000 &= \frac{\pi}{4} (d_c)^2 \times 100 \\
 &= 78.55 (d_c)^2 \\
 (d_c)^2 &= \frac{80000}{78.55} = 1018.5 \\
 d_c &= 31.9 \text{ mm} \quad \Rightarrow 32 \text{ mm}
 \end{aligned}$$

2. Core Selection Note & Standard Thread Specs

Adopt $d_c = 38$ mm to pass combined torsional stress

$$d_c = 38 \text{ mm}$$

$$d_o = 46 \text{ mm}$$

$$p = 8 \text{ mm}$$

$$\begin{aligned} d &= \frac{d_o + d_c}{2} \\ &= \frac{46 + 38}{2} \\ &= \mathbf{42 \text{ mm}} \end{aligned}$$

3. Helix Angle (α) & Thread Torque (T_1)

Torque to rotate screw against thread friction

$$\tan \alpha = \frac{p}{\pi \cdot d} = \frac{8}{\pi \times 42} = 0.0606$$

$$\begin{aligned} T_1 &= W \cdot \tan(\alpha + \varphi) \cdot \frac{d}{2} \\ &= 80000 \times \left[\frac{0.0606 + 0.14}{1 - (0.0606)(0.14)} \right] \times 21 \\ &= 80000 \times \left[\frac{0.2006}{0.9915} \right] \times 21 \\ &= \mathbf{340 \times 10^3 \text{ N} \cdot \text{mm} \quad (\text{Ans.})} \end{aligned}$$

4. Direct Compressive & Torsional Shear Stresses

(σ_c, τ)

Stress components in spindle body

$$\sigma_c = \frac{W}{\frac{\pi}{4}(d_c)^2} = \frac{80000}{\frac{\pi}{4}(38)^2} = 70.53 \text{ N/mm}^2$$

$$\tau = \frac{16T_1}{\pi(d_c)^3} = \frac{16 \times 340 \times 10^3}{\pi(38)^3} = 31.55 \text{ N/mm}^2$$

5. Maximum Principal Compressive Stress

$(\sigma_{c(\max)})$

Combined stress evaluation for allowable $\sigma_c =$
100 MPa

$$\begin{aligned} \sigma_{c(\max)} &= \frac{\sigma_c}{2} + \frac{1}{2} \sqrt{\sigma_c^2 + 4\tau^2} \\ &= 35.265 + \frac{1}{2} \sqrt{(70.53)^2 + 4(31.55)^2} \\ &= 35.265 + 47.315 \\ &= \mathbf{82.58 \text{ MPa}} \leq 100 \text{ MPa} \quad (\mathbf{SAFE}) \end{aligned}$$

6. Maximum Shear Stress ($\tau_{\max}$)

Maximum shear stress check for allowable $\tau =$
60 MPa

$$\begin{aligned} \tau_{\max} &= \frac{1}{2} \sqrt{\sigma_c^2 + 4\tau^2} \\ &= \frac{1}{2}(94.63) \\ &= \mathbf{47.32 \text{ MPa}} \leq 60 \text{ MPa} \quad (\mathbf{SAFE}) \end{aligned}$$

7. Spindle Buckling Check (W_{cr})

J.B. Johnson formula for unsupported length $L = 440$ mm

$$L = H_1 + \frac{h}{2} = 400 + 40 = 440 \text{ mm}, \quad k = 0.25d_c = 9.5 \text{ mm}$$

$$\begin{aligned} W_{cr} &= A_c \cdot \sigma_y \left[1 - \left(\frac{\sigma_y}{4\pi^2 EC} \right) \left(\frac{L}{k} \right)^2 \right] \\ &= 1134.1 \times 200 \left[1 - \left(\frac{200}{4\pi^2 (2.1 \times 10^5)(0.25)} \right) \left(\frac{440}{9.5} \right)^2 \right] \\ &= 226852 \times [1 - 0.207] = \mathbf{179.89 \text{ kN}} > 8 \end{aligned}$$

8. Thread Count (n) & Nut Height (h)

Bearing pressure equation ($p_b = 18 \text{ N/mm}^2$)

$$18 = \frac{80000}{\frac{\pi}{4}(46^2 - 38^2) \cdot n} = \frac{80000}{527.78n}$$

$$n = \frac{80000}{18 \times 527.78} = 8.4 \quad \Rightarrow \mathbf{n = 10 \text{ threads}}$$

$$h = n \cdot p = 10 \times 8 = \mathbf{80 \text{ mm}} \quad (\text{Ans.})$$

9. Thread Shear Stress in Screw (τ_{screw})

Shear check on screw threads ($t = \frac{p}{2} = 4$ mm)

$$\begin{aligned} \tau_{\text{screw}} &= \frac{W}{\pi \cdot d_c \cdot t \cdot n} \\ &= \frac{80000}{\pi \times 38 \times 4 \times 10} \\ &= \mathbf{16.15 \text{ MPa}} \leq 60 \text{ MPa} \quad (\text{SAFE}) \end{aligned}$$

10. Thread Shear Stress in Nut (τ_{nut})

Shear check on nut threads

$$\begin{aligned}
 \tau_{\text{nut}} &= \frac{W}{\pi \cdot d_o \cdot t \cdot n} \\
 &= \frac{80000}{\pi \times 46 \times 4 \times 10} \\
 &= \mathbf{13.84 \text{ MPa}} \leq 40 \text{ MPa} \quad (\mathbf{SAFE})
 \end{aligned}$$

11. Nut Body Outer Diameter (D_1)Tearing of nut body under allowable tension $\sigma_{t(\text{nut})} = 50 \text{ MPa}$

$$\begin{aligned}
 W &= \frac{\pi}{4}(D_1^2 - d_o^2) \cdot \sigma_{t(\text{nut})} \\
 80000 &= \frac{\pi}{4}(D_1^2 - 46^2) \times 50 = 39.3(D_1^2 - 2116) \\
 D_1^2 - 2116 &= \frac{80000}{39.3} = 2036 \\
 D_1 &= \sqrt{4152} = \mathbf{65 \text{ mm}} \quad (\mathbf{Ans.})
 \end{aligned}$$

12. Nut Collar Outer Diameter (D_2)

Crushing of collar under allowable compression

 $\sigma_{c(\text{nut})} = 45 \text{ MPa}$

$$\begin{aligned}
 W &= \frac{\pi}{4}(D_2^2 - D_1^2) \cdot \sigma_{c(\text{nut})} \\
 80000 &= \frac{\pi}{4}(D_2^2 - 65^2) \times 45 = 35.3(D_2^2 - 4225) \\
 D_2^2 - 4225 &= \frac{80000}{35.3} = 2266 \\
 D_2 &= \sqrt{6491} = 80.6 \text{ mm} \quad \Rightarrow \mathbf{D_2 = 82 \text{ mm}}
 \end{aligned}$$

13. Nut Collar Thickness (t_1)

Shearing of collar under allowable shear $\tau_{\text{nut}} = 40 \text{ MPa}$

$$W = \pi \cdot D_1 \cdot t_1 \cdot \tau_{\text{nut}}$$

$$80000 = \pi \times 65 \times t_1 \times 40 = 8170t_1$$

$$t_1 = \frac{80000}{8170} = 9.8 \text{ mm} \quad \Rightarrow t_1 = 10 \text{ mm} \quad (\text{Ans.})$$

14. Swivel Cup Head & Pin Diameters (D_3, D_4)

Empirical proportions for cup bearing

$$D_3 = 1.75d_o = 1.75 \times 46 = 80.5 \text{ mm} \quad \Rightarrow D_3 = 82 \text{ mm} \quad (\text{Ans.})$$

$$D_4 = 20 \text{ mm} \quad (\text{Loose Fit Pin in Cup})$$

15. Swivel Cup Collar Friction Torque (T_2)

Uniform pressure assumption between $R_1 = 41 \text{ mm}$, $R_2 = 10 \text{ mm}$

$$T_2 = \frac{2}{3} \mu_1 W \left[\frac{R_1^3 - R_2^3}{R_1^2 - R_2^2} \right]$$

$$= \frac{2}{3} (0.14) (80000) \left[\frac{41^3 - 10^3}{41^2 - 10^2} \right]$$

$$= 7.47 \times 10^3 \times \left[\frac{67921}{1581} \right]$$

$$= 321 \times 10^3 \text{ N} \cdot \text{mm} \quad (\text{Ans.})$$

16. Total Operating Torque (T) & Handle Length (L)Torque sum and operator effort $F = 300$ N

$$T = T_1 + T_2 = 340 \times 10^3 + 321 \times 10^3 = 661 \times 10^3 \text{ N} \cdot \text{mm}$$

$$L' = \frac{T}{F} = \frac{661 \times 10^3}{300} = 2203 \text{ mm}$$

$$L = 2250 \text{ mm} \quad (\text{Includes Gripping Allowance})$$

17. Handle Diameter (D) & Head Height (H)Handle bending stress limit $\sigma_b = 100$ MPa

$$M = F \cdot L = 300 \times 2250 = 675 \times 10^3 \text{ N} \cdot \text{mm}$$

$$675 \times 10^3 = \frac{\pi}{32} D^3 \times 100 = 9.82 D^3$$

$$D^3 = \frac{675 \times 10^3}{9.82} = 68.74 \times 10^3$$

$$D = 40.96 \text{ mm}$$

$$\Rightarrow D = 42 \text{ mm}$$

$$H = 2D = 2 \times 42 = 84 \text{ mm} \quad (\text{Ans.})$$

18. Ideal Torque (T_0) & Screw Jack Efficiency (η)

Efficiency evaluation relative to frictionless torque

$$T_0 = W \cdot \tan \alpha \cdot \frac{d}{2} = 80000 \times 0.0606 \times 21 = 101808 \text{ N} \cdot \text{mm}$$

$$\eta = \frac{T_0}{T}$$

$$= \frac{101808}{661 \times 10^3}$$

$$= 0.154 = 15.4\% \quad (\text{Ans.})$$

19. Body Inside Top Diameter (D_5)

Empirical relation for nut clearance

$$D_5 = 1.5D_2 = 1.5 \times 82 = \mathbf{123 \text{ mm}} \quad (\text{Ans.})$$

20. Body Wall Thickness (t_3)

Empirical relation for cast iron body wall

$$t_3 = 0.25d_o = 0.25 \times 46 = 11.5 \text{ mm} \Rightarrow t_3 = \mathbf{12 \text{ mm}} \quad (\text{Ans.})$$

21. Body Inside Bottom Diameter (D_6)

Empirical relation for tapered body

$$D_6 = 2.25D_2 = 2.25 \times 82 = \mathbf{185 \text{ mm}} \quad (\text{Ans.})$$

22. Body Outer Base Flange Diameter (D_7)

Empirical relation for base stability

$$D_7 = 1.75D_6 = 1.75 \times 185 = \mathbf{320 \text{ mm}} \quad (\text{Ans.})$$

23. Base Flange Thickness (t_2)

Empirical relation for base flange strength

$$t_2 = 2t_1 = 2 \times 10 = \mathbf{20 \text{ mm}} \quad (\text{Ans.})$$

24. Total Body Height (H_{body})

Height for lift, nut, and clearance

$$H_{\text{body}} = H_1 + h + 100 = 400 + 80 + 100 = \mathbf{580 \text{ mm}} \quad (\text{Ans.})$$

25. Complete Design Output

Screw Jack Summary: Spindle ($d_o = 46 \text{ mm}$, $d_c = 38 \text{ mm}$, $p = 8 \text{ mm}$), Nut ($h = 80 \text{ mm}$, $D_1 = 65 \text{ mm}$, $D_2 = 82 \text{ mm}$, $t_1 = 10 \text{ mm}$), Handle ($D = 42 \text{ mm}$, $L = 2250 \text{ mm}$), Body ($D_5 = 123$

**mm, D6 = 185 mm, D7 = 320 mm, H_body = 580 mm),
Efficiency eta = 15.4%**

SECTION 1: Screw Jack Complete Assembly — MECHANICAL DIAGRAM

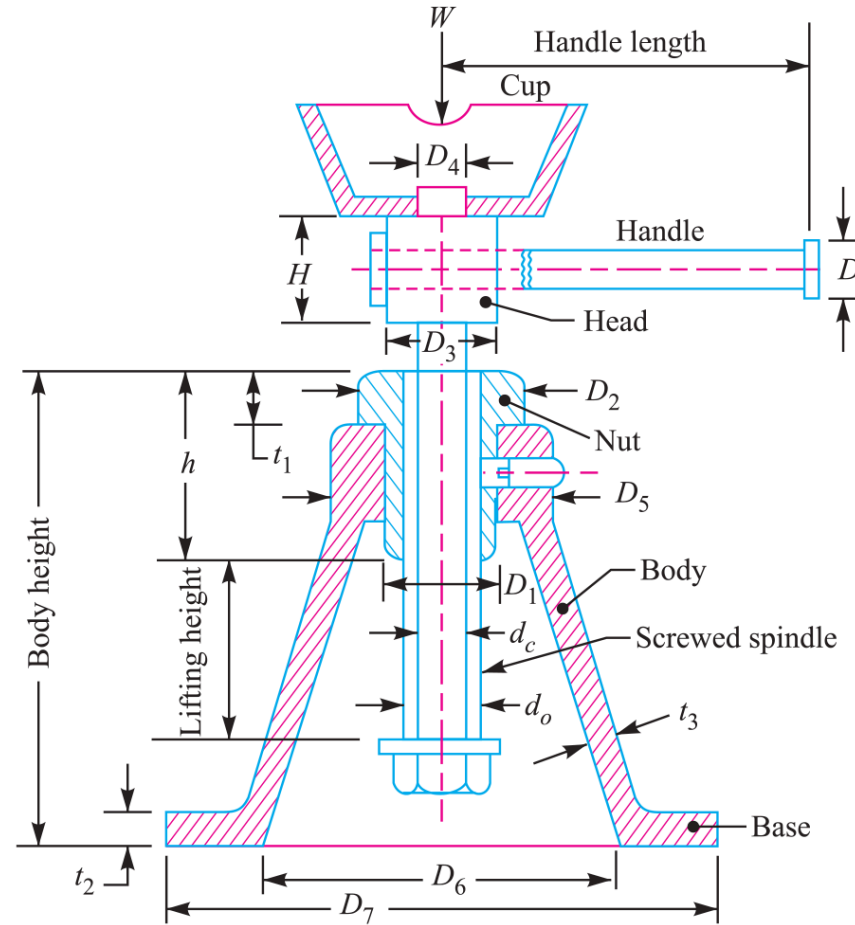


Fig. 17.11. Screw jack.

Figure 17.11: Complete Front Sectional Assembly of Screw Jack & All Component Proportions

SECTION 2: Differential Screw Jack Efficiency & Load Capacity Analysis

System Parameters

Nominal Diameter: $d_o = 50\text{ mm}$ | Thread Friction: $\mu = \tan \varphi = 0.15$

Upper Screw Pitch: $p_1 = 16\text{ mm}$ | Lower Screw Pitch: $p_2 = 12\text{ mm}$

Allowable Maximum Shear Stress: $\tau_{\max} = 28\text{ MPa}$

Design Protocol

Part 1: Calculate mean diameters (d_1, d_2) and helix angles (α_1, α_2)

Part 2: Calculate upper torque T_1 , lower torque T_2 , total torque T , & efficiency η

Part 3: Calculate upper screw core diameter d_{c1} , stresses (σ_c, τ), & max load capacity $W_{\max}$

1. Mean Diameters (d_1, d_2) & Helix Angles (α_1, α_2)

Upper & lower square thread geometry for $d_o = 50\text{ mm}$

$$d_1 = d_o - \frac{p_1}{2} = 50 - 8 = 42\text{ mm}$$

$$d_2 = d_o - \frac{p_2}{2} = 50 - 6 = 44\text{ mm}$$

$$\tan \alpha_1 = \frac{p_1}{\pi \cdot d_1} = \frac{16}{\pi \times 42} = 0.1212$$

$$\tan \alpha_2 = \frac{p_2}{\pi \cdot d_2} = \frac{12}{\pi \times 44} = 0.0868$$

2. Upper & Lower Thread Torques (T_1, T_2)Thread friction torque per unit load W

$$T_1 = W \left[\frac{0.1212 + 0.15}{1 - (0.1212)(0.15)} \right] \times 21 = 5.8W$$

$$T_2 = W \left[\frac{0.0868 - 0.15}{1 + (0.0868)(0.15)} \right] \times 22 = -1.37W$$

3. Total Operating Torque (T) & Ideal Torque (T_0)

Differential torque equations

$$T = T_1 - T_2 = 5.8W - (-1.37W) = 7.17W$$

$$T_0 = W \left[\frac{p_1 - p_2}{2\pi} \right] = \frac{4W}{2\pi} = 0.636W$$

4. Differential Screw Jack Efficiency (η)

Ratio of ideal torque to total operating torque

$$\eta = \frac{T_0}{T}$$

$$= \frac{0.636W}{7.17W}$$

$$= 0.0887 = \mathbf{8.87\%} \quad (\mathbf{Ans.})$$

5. Upper Screw Core Diameter (d_{c1}) & Stresses

Direct compression and torsional shear in critical upper screw

$$d_{c1} = d_o - p_1 = 50 - 16 = 34 \text{ mm}$$

$$\sigma_c = \frac{W}{\frac{\pi}{4}(34)^2} = 1.101 \times 10^{-3} WN/\text{mm}^2$$

$$\tau = \frac{16(5.8W)}{\pi(34)^3} = 0.7513 \times 10^{-3} WN/\text{mm}^2$$

6. Maximum Shear Stress Limit ($\tau_{\max} = 28 \text{ MPa}$)

Maximum shear stress equation

$$\tau_{\max} = \frac{1}{2} \sqrt{\sigma_c^2 + 4\tau^2}$$

$$28 = \frac{1}{2} \sqrt{(1.101 \times 10^{-3}W)^2 + 4(0.7513 \times 10^{-3}W)^2}$$

$$= 1.863 \times 10^{-3} \cdot W$$

7. Maximum Lifiable Load Capacity ($W_{\max}$)

Maximum safe load limit

$$W = \frac{28}{1.863 \times 10^{-3}}$$

$$= 30060 \text{ N}$$

$$= 30.06 \text{ kN} \quad (\text{Ans.})$$

8. Design Output

Differential Screw Jack: Efficiency $\eta = 8.87\%$, Max Load Capacity $W_{\max} = 30.06 \text{ kN}$ for $\tau_{\max} = 28 \text{ MPa}$

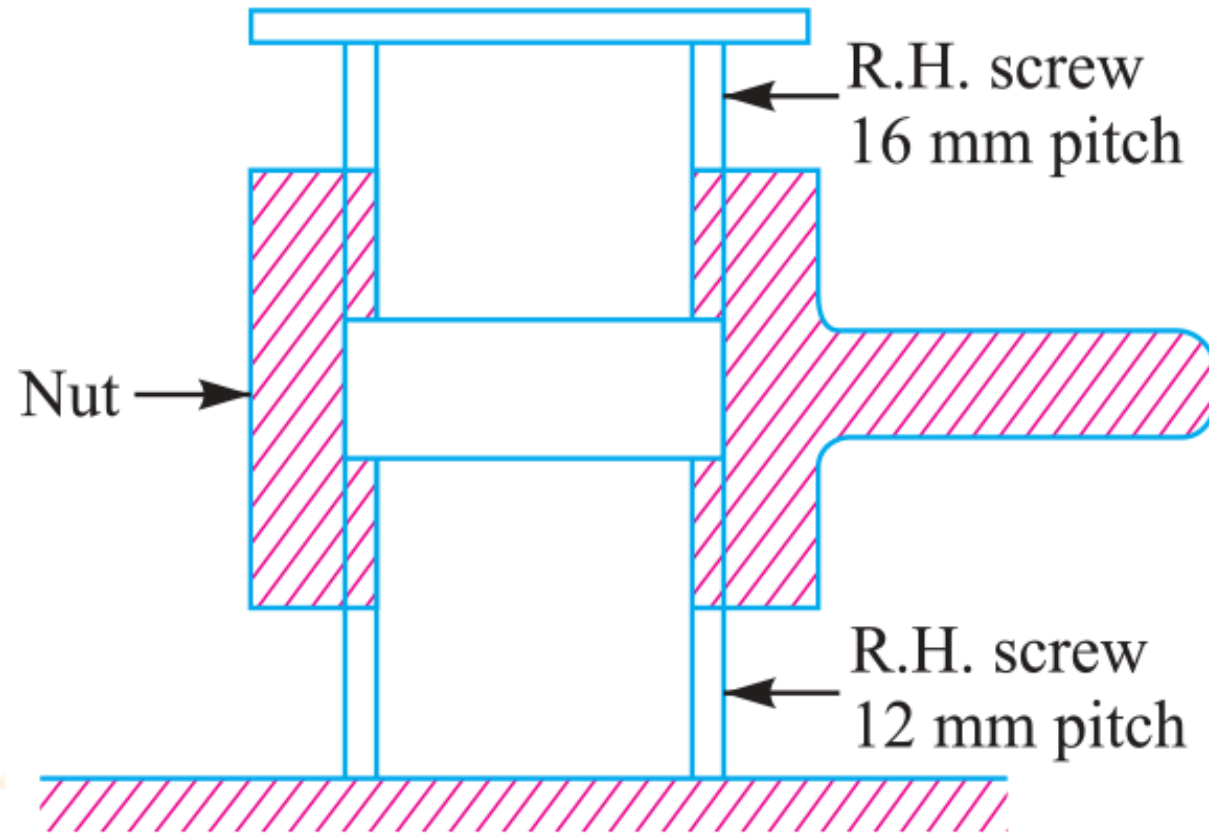
SECTION 2: Differential Screw Jack Assembly — MECHANICAL DIAGRAM

Fig. 17.16. Differential screw.

Figure 17.16: Differential Screw Jack Kinematic Layout & Thread Differential

SECTION REF 1: Table 17.1 — Standard Square Thread Dimensions (IS: 4694 – 1968)

Nominal Dia d (mm)	Major Dia d_1 (mm)	Minor Dia Bolt d_c (mm)	Pitch p (mm)	Thread Depth h (mm)	Thread Depth Nut H (mm)	Core Area A_c (mm ²)
10	10.0	8.0	2.0	1.0	1.25	50.3
12	12.0	10.0	2.0	1.0	1.25	78.5
14	14.0	12.0	2.0	1.0	1.25	113.0
16	16.0	14.0	2.0	1.0	1.25	154.0
18	18.0	16.0	2.0	1.0	1.25	201.0
20	20.0	18.0	2.0	1.0	1.25	254.0
22	22.0	19.0	3.0	1.5	1.75	284.0
24	24.0	21.0	3.0	1.5	1.75	346.0
26	26.0	23.0	3.0	1.5	1.75	415.0
28	28.0	25.0	3.0	1.5	1.75	491.0
30	30.0	27.0	3.0	1.5	1.75	573.0
32	32.0	29.0	3.0	1.5	1.75	661.0
(34)	34.0	31.0	3.0	1.5	1.75	755.0
36	36.0	33.0	3.0	1.5	1.75	855.0
(38)	38.0	35.0	3.0	1.5	1.75	962.0
40	40.0	37.0	3.0	1.5	1.75	1075.0
42	42.0	39.0	3.0	1.5	1.75	1195.0
44	44.0	41.0	3.0	1.5	1.75	1320.0
(46)	46.0	43.0	3.0	1.5	1.75	1452.0
48	48.0	45.0	3.0	1.5	1.75	1590.0
50	50.0	47.0	3.0	1.5	1.75	1735.0

Standard Specification: Dimensions per IS: 4694 – 1968 (Reaffirmed 1996) Fine Series square threads for power transmission.